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RESEARCH

The current state of artificial intelligence research in pediatric radiology and recommendations for the future: a scoping review

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Abstract

Background Most artificial intelligence (AI) research in radiology has focused on adults. Understanding macro-level trends in pediatric radiology AI can help guide, streamline, and bolster future research.

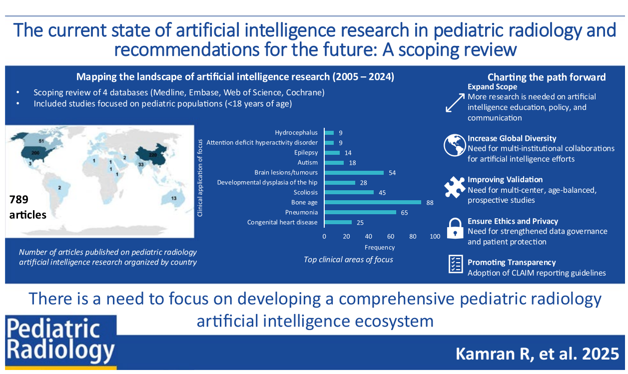
Objective To detail the current landscape of published AI research in pediatric radiology, filling a key research gap, as most radiology AI research has focused on adults.

Materials and methods We conducted a scoping review, with a comprehensive literature search of Medline, Embase, Web of Science, and Cochrane Library from 2005 to 2024. Literature included for review were (1) original articles, (2) investigations that focused on pediatric populations (<18 years of age), and (3) articles with direct applications to clinical radiology and AI. We extracted each article's study information, clinical application of focus, imaging modality, and the use of AI. We used descriptive frequencies to analyze summary statistics, and Chi-square testing to determine differences between categories.

Results In total, we found 4,376 articles and included 789 articles in the review. The top three countries most active in scholarship related to AI in pediatric radiology were China (220, 27.9%), the USA (200, 25.4%), and Canada (51, 6.5%) (P<0.001). The most common imaging modalities were radiography (298, 37.8%), MRI (260, 33.0%), and ultrasonography (114, 14.4%) (P<0.001). The most common subspecialties represented were musculoskeletal (260, 33.0%), neurological (227, 28.8%), and chest imaging (130, 16.5%) (P<0.001). The top two image analysis tasks discussed were image interpretation/diagnosis (719, 91.1%), and artifact and motion reduction/enhancing image quality (44, 5.6%) (P<0.001).

Conclusion Most pediatric radiology AI research originated from China and the USA, and focused on image interpretation/diagnosis. Thematic imbalances, particularly underrepresentation in research on communication, education, policy, and stakeholder perspectives, offer a guide for pediatric radiology AI development. There is a need for improved global collaboration and improved patient representativeness in datasets for pediatric radiology AI research to reduce bias with AI algorithms. The results from this scoping review offer a practical roadmap to inform future research and funding priorities in pediatric radiology AI.

Graphical Abstract



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Keywords Artificial intelligence · Pediatric radiology · Scoping review · Children · Radiography · Ultrasound · MRI · CT · Nuclear medicine · Imaging interpretation/diagnosis · Research

Introduction

Artificial intelligence (AI) is broadly defined as a technology that automates a cognitive task, such as machines simulating learning, comprehension, problem-solving, and decision-making [1]. Over the past few years, there has been an exponential growth of AI research in radiology. The many applications of the developed AI models range from performing image-recognition tasks [2, 3], scheduling medical imaging appointments and protocoling [4], to improving the overall efficiency of radiologists [5, 6]. As of January 2025, there were 758 Food and Drug Administration (FDA)-approved AI-based software devices for radiology, representing over 70% of all AI clinical applications' clearances; however, only 9% target pediatric imaging [2].

AI research in radiology has largely ignored the pediatric population, perhaps due to fewer pediatric imaging datasets available for AI training [7, 8], or the more complex pathologies related to the dynamic growth and development of children [9, 10]. In many instances, AI software designed for adults has been applied to children without modification [8]. This is problematic as pediatric radiology has its own unique set of challenges and difficulties (e.g., medicolegal and ethical considerations pertaining to imaging, rare and age-specific disease entities, developmental and anatomical variations, and structural alterations related to growth and development), and directly applying AI models developed for adults to children may lead to inconsistencies and inaccuracies [11]. It has also been reported that some AI models may have unpredictable impacts on patients. For example, AI models may generate inappropriate decisions, infringe on patient privacy, contribute to health inequities, or underperform in real clinical settings when compared to idealized or artificial environments created for research publications [12]. Pediatric radiologists need to be cognizant of these pitfalls when implementing AI applications with the goal of minimizing potential harm to children [12–15]. Furthermore, the current AI research in pediatric radiology mainly focuses on a few specific clinical areas, neglecting others. We lack a comprehensive, high-level examination of the existing scholarship on AI applications in pediatric radiology across different clinical areas.

This scoping review aims to map out the current literature on AI applications in pediatric radiology. Results from this review can be used to identify areas where more research is needed, and to identify major trends in AI research. By understanding macro-level trends in AI research in pediatric radiology, we can make concerted and targeted efforts to advance AI research in areas with knowledge deficiency.

Methods

This scoping review follows the Preferred Reporting Items for Systematic Reviews and Meta Analysis Extension for Scoping Reviews reporting guideline [16], and was prospectively registered on the Open Science Framework [17]. This study was reviewed by the Research Governance, Ethics, and Assurance Team at the University of Oxford and deemed exempt from ethical approval as this review did not collect participant human data.

Eligibility criteria

We included articles that met all the following eligibility criteria: (1) an original article; (2) focused on pediatric populations (or included a subgroup of pediatric populations where the mean/median age of the population was < 18 years of age [18]); (3) with a direct application to clinical radiology (e.g., diagnosis, clinical decision-making, administration, workflow, intervention, or clinical education); and (4) focused on an application of AI. Articles were excluded if they were not written in English, or constituted a review article, commentary, letter, editorial, conference abstract, or case series/case report.

Search strategy and selection process

Four databases: Medline, Embase, Web of Science, and the Cochrane Library were searched from January 1, 2005, to August 5, 2024. The reason for not searching the databases prior to 2005 is because of the limited AI research in radiology > 20 years ago [19]. There were no restrictions on the country of origin. The search strategy, developed in consultation with an information scientist, is detailed in the Supplementary Material. Articles were screened for title, abstract and full-text independently, in duplicate, and sequentially, on the Covidence review management platform (R.K., C.C., N.V., S.L., L.J., Y.S., L.L., A.C.L.) with disagreements resolved by a third reviewer (R.K.). Eligible articles underwent data extraction in duplicate (R.K., A.L., R.Y., A.S., L.C., J.B., C.C., N.V., S.L., L.J., Y.S., L.L., A.C.L.), with disagreements similarly resolved by a third reviewer (R.K.). All researchers involved hold doctoral-level degrees (e.g., MD, PhD, DPhil), or a research-based master's degree (e.g., MSc), and all authors had prior experience in systematic and scoping reviews.

Data items

Extracted data included the following: study information (title of article, author, year of publication, country of corresponding author); sub-specialty of focus (e.g., cardiac, neurological, musculoskeletal, nuclear medicine, chest, body/abdominal, interventional, head/neck, and/or other, with the flexibility to have more than one focus per study); clinical application of focus; imaging modality; how AI was used, category of technology (e.g., deep learning), and if bias/limitations of the AI model were discussed. The use of AI was categorized according to predefined categories, as described in a white paper by Tang and colleagues on behalf of the Canadian Association of Radiologists on AI applications in radiology [20]. The data categories were finalized based on iterative meetings held with the Pediatric Research Subcommittee of the American College of Radiology, where the data extraction template underwent revision. For data extraction, a consensus meeting was held with researchers involved with screening/data extraction to ensure consistency in understanding categories and with categorizing articles. The author, subspecialty/training level, and years of clinical experience, as relevant, are as follows: R.K., diagnostic radiology resident, 1 year; E.W., pediatric neuroradiologist, 18 years; A.S., medical student, 2 years; J.B., medical student, 2 years; L.C., medical student, 2 years; A.L., medical student, 2 years; Y.X.J., medical student, 2 years; C.H.C., family physician, 4 years; N.V., graduate student, 3 years; S.L., medical student, 4 years; L.J., medical student, 4 years; Y.S., medical student, 3 years; L.L., graduate student, 2 years; A.C.L., general pediatrics resident, 1 year; G.K., pediatric radiologist, 21 years; A.T.T., pediatric radiologist and nuclear radiologist, 12 years; M.B.K.S., pediatric radiologist, 16 years; R.K.O., pediatric radiologist, 35 years; M.G., pediatric radiologist, 16 years; C.M., pediatric radiologist, 7 years; M.L.H., pediatric neuroradiologist, 10 years; M.G., medical student, 3 years; H.O., pediatric radiologist, 11 years; S.R.T., pediatric radiologist and pediatric neuroradiologist, 5 years; M.A.B., pediatric radiologist, musculoskeletal radiology and fetal imaging, 4 years; A.T., pediatric musculoskeletal radiologist, 15.5 years; S.A., pediatric radiologist and pediatric neuroradiologist, 26 years; S.C., pediatric radiology, 11 years; A.S.D., pediatric radiologist, 23 years.

Synthesis methods and statistical analysis

We used frequencies to describe the extracted data, and Chi-square testing to determine whether the frequencies in each category were significantly different from each other. A P-value of <0.05 represented statistical significance in this study. We analyzed the data on IBM SPSS version 28.0 (Chicago, IL). To support the interpretation of findings and the development of meaningful recommendations, we held four roundtable-style meetings via Zoom over the course of 1 year with the author team. These meetings were used to discuss results, contextualize key trends, identify gaps in the literature, and collaboratively shape the recommendations. Authors brought multidisciplinary perspectives to the table, including expertise in pediatric radiology, artificial intelligence, implementation science, and data ethics. In addition to live discussions, we maintained a shared Google document throughout the drafting process for recommendations to be embedded as part of the discussion of this scoping review. This Google document served as a collaborative workspace where all authors could contribute to, edit, and iteratively refine the recommendations.

Results

Through our systematic search, we identified 4,376 articles and included 789 articles (Supplementary Material) in this scoping review (Fig. 1).

Demographic trends in pediatric radiology artificial intelligence research

The number of AI-related publications exponentially increased since 2017 (Fig. 2). The top five countries most active in contributing to scholarship related to AI applications in pediatric radiology were as follows: China (220, 27.9%), the USA (200, 25.4%), Canada (51, 6.5%), South Korea (44, 5.6%), and India (33, 4.2%) (Fig. 3). A total of 149 (18.9%) articles were published in computer science/engineering/physics journals, while 640 (81.1%) articles were published in journals with a clinical/medical focus. Table 1 provides an overview of the year of publication of the included articles, with the Supplementary Material providing a detailed overview of all countries represented in this scoping review.

Imaging modalities covered by pediatric radiology artificial intelligence research

The following imaging modalities were represented in pediatric radiology AI research: computed tomography (CT), magnetic resonance imaging (MRI), nuclear medicine scans, radiography, ultrasonography (US), and optical coherence tomography (OCT). Table 2 provides an overview of the imaging modalities used in the included articles, with the three most common modalities being radiography (298, 37.8%), MRI (260, 33.0%), and US (114, 14.5%).

Subspecialties, clinical applications of focus, category of technology used, and discussion of bias for pediatric radiology artificial intelligence research

Of the included articles, the three most common clinical subspecialties were musculoskeletal (260, 33.0%), neurological (227, 28.8%), and chest (130, 16.5%) (Table 3). The three most common clinical applications were bone age estimation (88, 11.2%), pneumonia detection (65, 8.2%), and brain lesions/tumor identification (54, 6.8%). Figure 4 displays the top ten clinical applications of AI in pediatric radiology and the imaging modalities used for each specific disease, by number of articles.

Use of artificial intelligence in pediatric radiology

The most common use of AI in pediatric radiology research was in image interpretation/diagnosis (719, 91.1%), followed by artifact and motion reduction/enhancing image quality (44, 5.6%). Table 4 provides an overview of the current applications of AI in pediatric radiology within the articles included in this review, by number of articles. This data is also graphically presented in Fig. 5. Most articles focused on non-radiomics-based analyses (712, 90.2%), with 72 articles (9.1%) focusing on radiomics-based analyses, and 5 (0.6%) unspecified. A total of 701 (88.8%) of the articles discussed biases of AI/their AI model, with some articles (88, 11.2%) demonstrating reporting bias, specifically, selective or limited reporting of important methodological information related to their AI model (e.g., articles presented accuracy metrics but did not contextualize limitations in dataset composition, missing-data handling, or clinical workflow differences that could attenuate real-world performance, limiting interpretability). A total of 52 articles (6.6%) did not report the dataset used for AI algorithm development, while 511 articles (64.8%) relied on local hospital datasets. The remaining articles used a highly heterogeneous mix of databases, with limited overlap across publications.

Discussion

This review details the scope of the current AI research in pediatric radiology. It reveals that the number of AI-related publications increased substantially since 2017. Most of the AI research was conducted in China and the USA, with the most common clinical applications being bone age assessment, pneumonia detection, and brain tumor identification. Most articles (722, 91.5%) focused on image interpretation/diagnosis, with a paucity of research focusing on other applications of AI in pediatric radiology (e.g., communication, consultation, education, image acquisition, policy). This was a key finding from our study, as the areas where AI is underutilized in pediatric radiology (communication and education) are relatively simpler to implement compared to complex tasks like image interpretation or diagnosis. Yet, these opportunities are often overlooked. Addressing these gaps presents a promising avenue for AI to support and enhance pediatric radiology.

Several key insights and recommendations can be drawn from this study. Firstly, this scoping review included a large number of articles ($n = 789$), which demonstrated that although AI research in pediatric radiology is still lagging behind adult radiology [21–24], the amount of research is increasing [9].

Pediatric radiology is a unique clinical arena for AI research due to the complexities of the disease processes, the fragility of the patient population, and the sensitive nature of the available imaging data. Safeguards must be in place to prevent unintended harm caused by AI tools (e.g., AI-generated recommendations that lead to unnecessary imaging and increased radiation exposure, or inexperienced radiologists being influenced by inaccurate AI outputs) [12]. There is a critical need to carefully and thoughtfully design and implement pediatric-specific AI models to minimize their potential harm. Yet despite this need, the current review identified only a few articles on the topics of policy, communication, and education. Further, only one (0.1%) article within this review focused on gathering patient and key stakeholder perspectives on AI [25]. This gap in the literature may lead to several negative consequences: AI algorithms may not align with patient-centered care; adoption and implementation could be hindered by a lack of clinician and stakeholder buy-in; and there is a risk of misunderstanding, misusing AI tools in clinical settings, or introducing unintended biases and safety concerns. Without adequate research into policy frameworks, education, and stakeholder perspectives, the integration of AI in pediatric radiology risks failing to meet the unique needs of pediatric patients and could inadvertently compromise the quality of care.

This scoping review identifies that certain subspecialty areas and thematic domains, such as policy, education, and implementation science, are underrepresented in pediatric radiology AI research. This uneven distribution limits the field's ability to holistically evaluate and responsibly integrate AI into pediatric imaging practice. National and international radiology organizations should actively incentivize and support research in underexplored domains, including ethical frameworks, clinical integration pathways, communication strategies, and education and training for AI adoption.

We also identified a limited range of areas in pediatric radiology in which AI tools have been examined. While the range of clinical areas that can benefit from AI in pediatric radiology is expanding, there remains a significant imbalance in the frequency of articles across different clinical areas in pediatric radiology. We postulate several reasons for this disparity. Firstly, clinical areas that are attracting

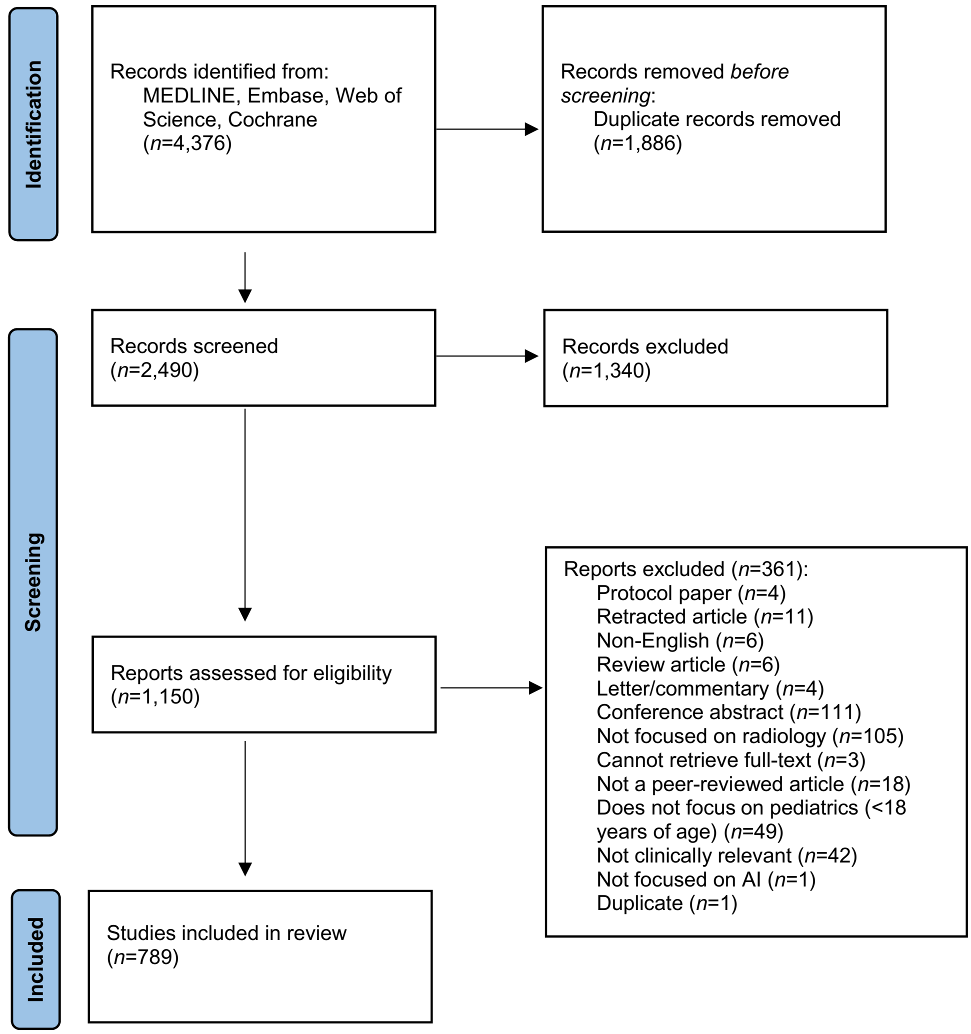

Fig. 1 PRISMA diagram of study selection

more attention from developers typically have easier access to datasets for AI model training and validation, along with more robust funding. Additionally, AI algorithms in these fields may be more straightforward to develop and implement clinically, serving as stepping-stones before addressing more complex tasks. Broadening the scope of AI research can be achieved by fostering sustained, interdisciplinary collaboration among pediatric radiologists, clinicians, computer scientists, and software engineers. Future efforts should not only target high-yield applications with well-defined outcomes, but also explore underrepresented and diagnostically complex areas, such as neonatal imaging, musculoskeletal anomalies, congenital disorders, and functional imaging.

This scoping review also demonstrates that potential biases in primary studies on AI in pediatric radiology are often not fully addressed in the published papers. Lack of transparency and failure to acknowledge bias hinder critical appraisal, limit reproducibility, and pose challenges for translating findings into clinical practice. To improve reporting quality and promote responsible AI development, authors should be encouraged to adhere to established reporting frameworks such as the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) guidelines when preparing manuscripts. Emphasis should be placed on the "Discussion" section, where authors should explicitly outline study limitations, including selection bias, dataset representativeness, statistical uncertainty, and the generalizability of results to real-world pediatric settings. Journals and peer reviewers also play a key role and should prioritize the enforcement of standardized reporting practices to enhance methodological rigor and transparency in pediatric AI research.

Below, key deficiencies in AI literature in pediatric radiology and potential solutions are presented. The recommendations below were developed through an iterative, collaborative process among the author team, drawing on diverse clinical and research expertise. Through a series of roundtable discussions and ongoing shared input, we formed a set of forward-looking recommendations to guide more balanced, inclusive, and impactful AI research in pediatric radiology linked to current deficiencies.

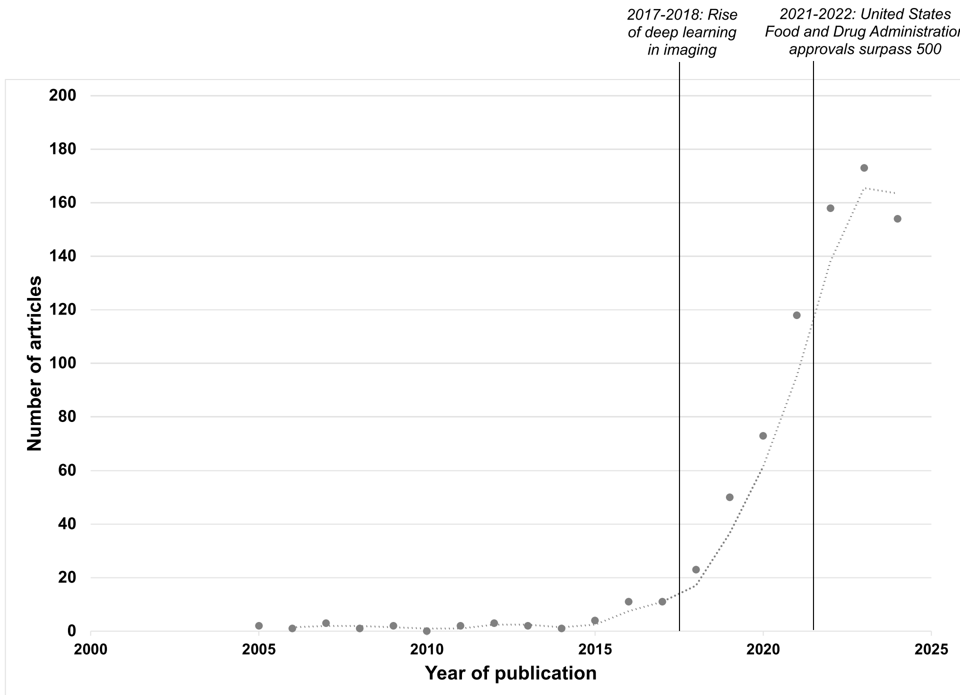

Fig. 2 Number of articles published on pediatric radiology artificial intelligence research by year

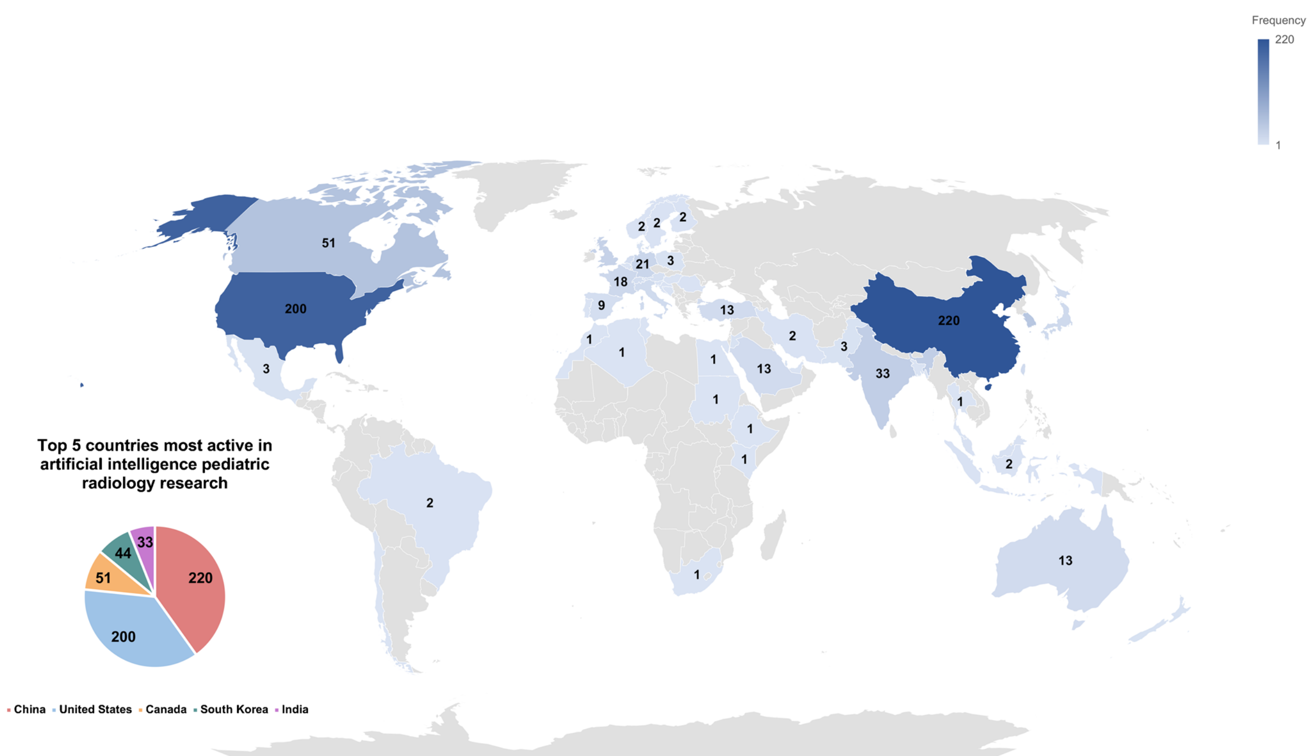

Fig. 3 Number of articles published on pediatric radiology artificial intelligence research organized by country

Table 1 Yearly frequency of the included articles of this scoping review

Year	Frequency (number of articles, *n*)	Percent	*P*-value
2005	2	0.3	**< 0.001**
2006	1	0.1	
2007	3	0.4	
2008	1	0.1	
2009	2	0.3	
2010	0	0.0	
2011	2	0.3	
2012	3	0.4	
2013	2	0.3	
2014	1	0.1	
2015	4	0.5	
2016	11	1.4	
2017	11	1.4	
2018	23	2.9	
2019	50	6.3	
2020	72	9.1	
2021	118	15.0	
2022	158	20.0	
2023	171	21.7	
2024	154	19.5	
Total	**789 articles**		

Limited datasets for pediatric artificial intelligence research

Recommendation: Many AI models are trained on small or institution-specific datasets, limiting their ability to generalize and undermining their clinical utility in real-world practice [8, 26–30]. We recommend the development of secure, collaborative data-sharing frameworks among pediatric healthcare institutions nationally and internationally. Multi-institutional consortia can enable the aggregation of anonymized, standardized imaging data across a broad spectrum of age groups, pathologies, and imaging modalities. Such efforts will enhance the representativeness and robustness of AI models, enabling them to perform more reliably in diverse clinical environments. Additionally, establishing clear governance, ethical safeguards, and data stewardship protocols will be essential to protect patient privacy while advancing innovation.

Table 2 Frequency of the imaging modalities represented among included articles

Modality	Frequency (number of articles, *n*)	Percent	*P*-value
Radiography	298	37.8	**< 0.001**
MRI	260	33.0	
US	114	14.5	
CT	90	11.4	
Nuclear	16	2.0	
General[a]	7	0.9	
OCT	4	0.5	
Total	**789 articles**		

CT computed tomography, *MRI* magnetic resonance imaging, *OCT* ocular coherence tomography, *US* ultrasound. [a]Articles were categorized as general if they did not focus on a subspecialty; e.g., if they focused on understanding stakeholders' perspectives of AI in pediatric radiology, education, or workflow optimization

Limited origin of available publications of artificial intelligence in pediatric radiology (most of the articles in this scoping review were published in China and the USA)

Recommendation: The fact that most articles included in this review originated from China and the USA raises concerns about the representativeness of datasets and the applicability/generalizability of AI tools to diverse healthcare settings and patient populations. Future research should explicitly acknowledge and address geographic and population-level biases embedded within existing AI datasets. These biases may impact diagnostic accuracy and exacerbate health inequities, particularly for under-represented populations [31, 32]. Specifically, high-income country–concentrated datasets risk domain shift, and miscalibration when these models are used elsewhere, including the potential for false-negatives when used in under-represented groups, and potentially over-calling findings where prevalence statistics may differ. To foster the development of more inclusive and equitable AI tools, global collaboration should be prioritized,

Table 3 Frequency of subspecialty focus represented among included articles

Subspecialty	Frequency (number of articles, *n*)	Percent	*P*-value
Musculoskeletal	260	33.0	**<0.001**
Neurological	227	28.8	
Chest	130	16.5	
Body/abdominal	110	13.9	
Cardiac	46	5.8	
Head/neck	9	1.1	
General/not applicable[a]	7	0.9	
Total	**789 articles**		

[a]Articles were categorized as general if they did not focus on a subspecialty; e.g., if they focused on understanding stakeholders' perspectives of AI in pediatric radiology, education, or workflow optimization

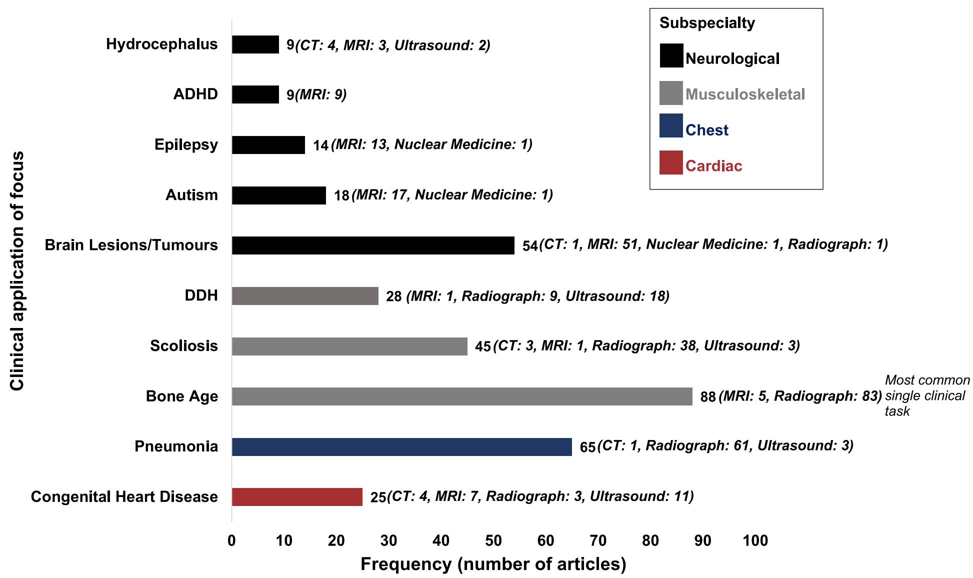

Fig. 4 Clinical application of focus and imaging modalities used for pediatric radiology artificial intelligence research articles. ADHD, attention-deficit/hyperactivity disorder; AI, artificial intelligence; CT, computed tomography; DDH, developmental dysplasia of the hip; MRI, magnetic resonance imaging

encouraging contributions from countries with limited representation in the current literature, ideally in partnership with institutions that possess robust AI infrastructure. Such efforts can support the creation of multinational, demographically diverse datasets, ultimately improving the generalizability, fairness, and clinical utility of AI applications in pediatric radiology [33].

Validation issues of artificial intelligence algorithms in pediatric radiology

Recommendation: Robust validation of AI algorithms is necessary to ensure the reliability, generalizability, and safety of AI models across varied clinical settings and patient populations. Given the heterogeneity of pediatric patients, including variations in age, anatomy, pathology presentation, and imaging protocols, AI models must be tested beyond their development datasets using multi-institutional, prospective, and demographically representative data. There is also somewhat sparse availability of normalized data (e.g., by age, gender, height, weight, BSA). Training pediatric radiology AI models remains a challenge due to the dynamic nature of growth and development. Anatomy and pathology may present very differently in a 5-year-old versus a 17-year-old, requiring large, image-specific datasets. For example, neonatal brains with smooth sulci and gyri differ markedly from adolescent brains. AI models do not always take this into consideration. This need for granular data across age groups, paired with the imbalance of

Table 4 Frequency and types of artificial intelligence (AI) applications in pediatric radiology [20]

Application of AI	Frequency (number of articles, *n*)	Percent	*P*-value
Image interpretation/diagnosis	719	91.1	**< 0.001**
Artifact/motion reduction/image processing and enhancement	44	5.6	
Image acquisition	8	1.0	
Communication	7	0.9	
Consultation (protocoling, appropriateness, review prior images)	4	0.5	
Education	3	0.4	
Image interpretation/diagnoses; policy making; communication; education	1	0.1	
Other: Discover relationships between selected patient features and no-show or long waiting room time endpoints	1	0.1	
Other: Understanding patient perspectives on artificial intelligence	1	0.1	
Policy	1	0.1	
Total	**789 articles**		

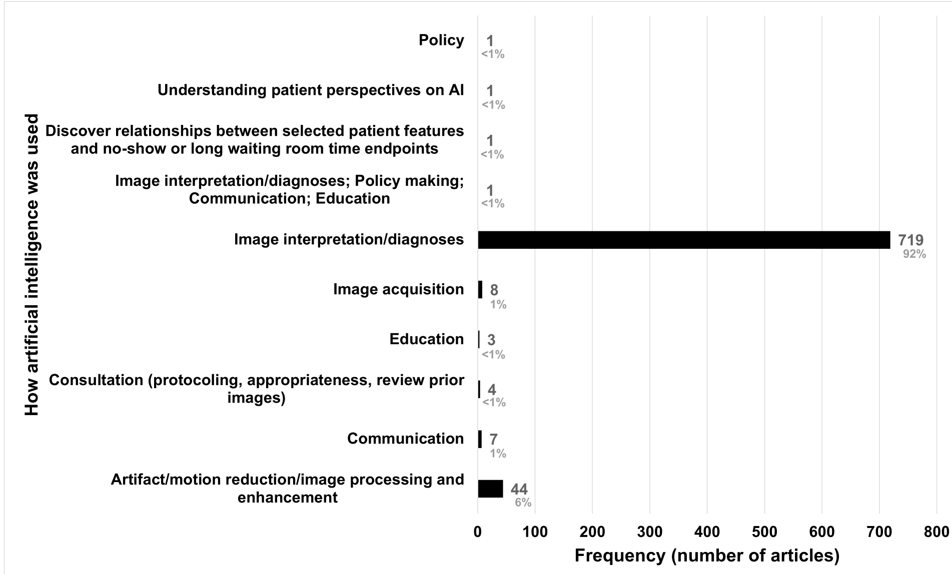

Fig. 5 Use of artificial intelligence in pediatric radiology research. AI, artificial intelligence

datasets between ages, poses difficulty in creating robust AI models in pediatric radiology. Children are also less frequently imaged than adults, often to minimize radiation risk; however, this creates a lack of longitudinal data and fewer scans. This restricts the ability to capture changes across age groups and adds to the challenge of developing robust models. Equally important is the continuous involvement of key stakeholders: radiologists, technologists, ethicists, patients and families, and institutional leaders, throughout the entire AI lifecycle. From initial development to testing, regulatory review, and implementation, stakeholder engagement helps to identify and address ethical concerns (e.g., bias, accountability, consent), enhance transparency, and promote clinical trust.

Regulatory considerations, data privacy, commercial incentives, and transparency in pediatric radiology artificial intelligence research

Recommendation: An important consideration in the development and clinical adoption of AI tools in pediatric radiology is regulatory oversight, particularly from agencies such as the U.S. Food and Drug Administration (FDA) in the USA. The FDA plays a central role not only in funding-related pathways but in determining how AI models are classified, reviewed, and approved for clinical use [2]. Articles relying on open-source datasets, while useful for algorithm development and benchmarking, raise important concerns around data quality, standardization, and especially patient privacy. The use of publicly available data may lack the clinical depth, demographic diversity, and ethical safeguards required for regulatory-grade AI tools. Future research should prioritize secure, ethically governed data acquisition, and should engage proactively with regulatory frameworks to ensure that emerging AI tools are not only innovative but also safe, equitable, and clinically viable. Another important consideration for future research is the commercial nature of AI development in radiology. Unlike traditional academic or public health research, many AI tools are developed by private entities with commercial incentives, which can influence publication practices and data transparency. Similar to drug development, there may be selective reporting, with articles published only when findings support the product's performance. However, unlike pharmaceuticals, there is often no regulatory requirement to publicly share validation data or negative results for non–FDA-approved tools. This creates a risk of publication bias and underreporting of limitations, particularly regarding dataset complexity, generalizability, and model performance. Future research, and potentially future policy, should encourage more transparent reporting standards [34] and incentives for sharing real-world performance data, even from commercial developers, to ensure the safe and equitable integration of AI into pediatric radiology.

Strengths and limitations

Our scoping review has strengths worth highlighting. We examined a large number of articles. Given the large sampling, we are confident that the information gathered reflects current clinical and demographic trends of AI research in pediatric radiology. Furthermore, we used the Canadian Association of Radiologists' guidelines to categorize AI applications [20]. This helped to create a consistent and standardized framework for discussing AI use in radiology. This approach enhances the generalizability of our findings to future reviews that may encompass both pediatric and adult radiology.

This review has limitations, which we should acknowledge as well. Despite conducting a comprehensive literature search, it is possible that some relevant articles were missed, which is always a risk when conducting this type of literature search. Exclusion of non-English articles may also disproportionately underrepresent Global South research. Results might include publication bias as we only included articles published in the English medical literature. Cohen's kappa was also not calculated for reviewer screening. This review does not provide any detailed information regarding the AI models, or their dataset size, validation strategies, or performance. This limitation impacts the evaluation of AI model robustness. This was intentional, to limit the scope and verbosity of the review. Our categorization of computational approaches (or technology used) revealed a limitation in how current pediatric AI articles report their methods. The classification scheme organized articles into radiomics-based and non-radiomics-based analyses, and was based on how authors reported their methods in included articles. This classification scheme is simplistic, and a limitation of this categorization method is that it does not capture the diversity of AI methods. There was marked heterogeneity in how authors reported their computational approaches, and this reiterates the usefulness of adhering to standardized reporting guidelines to better inform researchers about which computational approaches may be most effective for specific pediatric applications. We also did not extract or analyze the sample sizes of the datasets used in the included articles. As a result, we are unable to quantify the extent of this issue across the literature. This review also does not compare AI model accuracy, which can be explored by future systematic reviews.

Future work

Future research would benefit from granular systematic reviews focused on specific subsets of pediatric radiology AI research, particularly in areas with a reasonable number of primary articles (such as AI models aimed at reducing artifacts and enhancing images in pediatric radiology), as this scoping review did not investigate sub-group analyses. National and international radiology organizations should incentivize studies in underrepresented areas, including policy, education, communication, and stakeholder perspectives, which currently lack original research. Understanding the size and characteristics of datasets, particularly in comparison to those used in adult-focused AI studies, would provide valuable context for assessing the scalability and generalizability of pediatric AI models. Future reviews should consider systematically evaluating dataset sizes and comparing them to adult counterparts, to better characterize the data landscape and inform targeted efforts to address these limitations. The field would also greatly benefit from broader collaborative research efforts that bring together multidisciplinary teams, including radiologists, AI researchers, pediatricians, and data scientists. Such collaborations would support a more comprehensive exploration of AI algorithms specifically designed for pediatric care. By fostering these partnerships, we can advance the understanding of the unique challenges and opportunities associated with pediatric AI applications.

Call to action

Leaders in pediatric radiology, including pediatric radiologists, pediatric radiology journal editorial teams, pediatric radiology research funders, and commissioners, must act now to improve key areas of need for pediatric radiology AI research. Top priorities include (1) building and sharing representative, multi-institutional pediatric radiology datasets; (2) requiring transparent and standards-based reporting (e.g., CLAIM) with explicit limitations, subgroup results, and calibration; (3) redirecting funding and protected time towards unexplored areas such as policy, communication, education, and implementation, as well as validation of AI algorithms across ages, sites, and prevalence settings (e.g., high-, low-, and middle-income countries); (4) ensuring that thoughtful stakeholder engagement (e.g., children, families, interdisciplinary clinical team members, ethicists) is embedded across the AI lifecycle to ensure safety, equity, and real-world utility.

Conclusion

This scoping review provides radiologists and researchers with a broad overview of the current research landscape of AI in pediatric radiology. It identifies the field's strengths, weaknesses, and key areas in need within this research landscape. This information can serve as a valuable roadmap to guide and strengthen future AI research efforts in pediatric radiology.

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Declarations

Conflicts of interest None

References

1. Thodberg HH, Thodberg B, Ahlkvist J, Offiah AC (2022) Autonomous artificial intelligence in pediatric radiology: the use and perception of BoneXpert for bone age assessment. Pediatr Radiol 52:1338–1346. https://doi.org/10.1007/s00247-022-05295-w
2. Murphy H (2025 January) FDA has approved over 1,000 clinical AI applications, with most aimed at radiology. Health Imaging. Available from: https://healthimaging.com/topics/artificial-intelligence/fda-has-approved-over-1000-clinical-ai-applications-most-aimed-radiology
3. Hosny A, Parmar C, Quackenbush J, Schwartz LH, Aerts HJWL (2018) Artificial intelligence in radiology. Nat Rev Cancer 18:500–510. https://doi.org/10.1038/s41568-018-0016-5
4. Syed AB, Zoga AC (2018) Artificial intelligence in radiology: current technology and future directions. Semin Musculoskelet Radiol 22:540–545. https://doi.org/10.1055/s-0038-1673383
5. Jin D, Harrison AP, Zhang L, Yan K, Wang Y, Cai J, et al (2021) Chapter 14 - Artificial intelligence in radiology. In: Xing L, Giger ML, Min JK, editors. Artificial intelligence in medicine [Internet]. Academic Press; [cited 2024 Sep 29]. p. 265–89. Available from: https://www.sciencedirect.com/science/article/pii/B9780128212592000144
6. Thrall JH, Li X, Li Q, Cruz C, Do S, Dreyer K et al (2018) Artificial intelligence and machine learning in radiology: opportunities, challenges, pitfalls, and criteria for success. J Am Coll Radiol 15:504–508. https://doi.org/10.1016/j.jacr.2017.12.026
7. Shelmerdine SC, Rosendahl K, Arthurs OJ (2022) Artificial intelligence in paediatric radiology: international survey of health care professionals' opinions. Pediatr Radiol 52:30–41.https://doi.org/10.1007/s00247-021-05195-5
8. Davendralingam N, Sebire NJ, Arthurs OJ, Shelmerdine SC (2021) Artificial intelligence in paediatric radiology: future opportunities. Br J Radiol 94:20200975. https://doi.org/10.1259/bjr.20200975
9. Otjen JP, Moore MM, Romberg EK, Perez FA, Iyer RS (2022) The current and future roles of artificial intelligence in pediatric radiology. Pediatr Radiol 52:2065–2073. https://doi.org/10.1007/s00247-021-05086-9
10. Tierradentro-Garcia LO, Sotardi ST, Sammer MBK, Otero HJ (2023) Commercially available artificial intelligence algorithms of interest to pediatric radiology: the growing gap between potential use and data training. J Am Coll Radiol 20:748–751. https://doi.org/10.1016/j.jacr.2023.04.017
11. Alqahtani FF, Messina F, Offiah AC (2019) Are semi-automated software program designed for adults accurate for the identification of vertebral fractures in children? Eur Radiol 29:6780–6789. https://doi.org/10.1007/s00330-019-06250-4
12. Ciet P, Eade C, Ho ML, Laborie LB, Mahomed N, Naidoo J et al (2024Apr) The unintended consequences of artificial intelligence in paediatric radiology. Pediatr Radiol 1:585–593. https://doi.org/10.1007/s00247-023-05746-y
13. Offiah AC (2022) Current and emerging artificial intelligence applications for pediatric musculoskeletal radiology. Pediatr Radiol 52:2149–2158. https://doi.org/10.1007/s00247-021-05130-8
14. Desai SB, Pareek A, Lungren MP (2022) Current and emerging artificial intelligence applications for pediatric interventional radiology. Pediatr Radiol 52:2173–2177. https://doi.org/10.1007/s00247-021-05013-y
15. Meshaka R, Pinto Dos Santos D, Arthurs OJ, Sebire NJ, Shelmerdine SC (2022) Artificial intelligence reporting guidelines: what the pediatric radiologist needs to know. Pediatr Radiol 52:2101–2110. https://doi.org/10.1007/s00247-021-05129-1
16. Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D et al (2018Oct) PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. Ann Intern Med 2:467–473. https://doi.org/10.7326/m18-0850
17. Kamran R (2024 Aug 21) Artificial intelligence in pediatric radiology: a scoping review protocol organised according to the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) Checklist. [cited 2024 Sep 29]; Available from: https://osf.io/hjbmg/
18. Wang KC, Wang KC, Amirabadi A, Cheung E, Uleryk E, Moineddin R et al (2014) Evidence-based outcomes on diagnostic accuracy of quantitative ultrasound for assessment of pediatric osteoporosis — a systematic review. Pediatr Radiol 44:1573–1587. https://doi.org/10.1007/s00247-014-3041-x
19. Kibrom BT, Manyazewal T, Demma BD, Feleke TH, Kabtimer AS, Ayele ND et al (2024) Emerging technologies in pediatric radiology: current developments and future prospects. Pediatr Radiol 54:1428–1436. https://doi.org/10.1007/s00247-024-05997-3
20. Tang A, Tam R, Cadrin-Chênevert A, Guest W, Chong J, Barfett J et al (2018) Canadian association of radiologists white paper on artificial intelligence in radiology. Can Assoc Radiol J 69:120–135. https://doi.org/10.1016/j.carj.2018.02.002
21. Kocak B, Baessler B, Cuocolo R, Mercaldo N, Pinto dos Santos D (2023 Nov 1) Trends and statistics of artificial intelligence and radiomics research in radiology, nuclear medicine, and medical imaging: bibliometric analysis. Eur Radiol 33:7542–55. https://doi.org/10.1007/s00330-023-09772-0
22. van Leeuwen KG, Schalekamp S, Rutten MJCM, van Ginneken B, de Rooij M (2021) Artificial intelligence in radiology: 100 commercially available products and their scientific evidence. Eur Radiol 31:3797–3804. https://doi.org/10.1007/s00330-021-07892-z
23. Boeken T, Feydy J, Lecler A, Soyer P, Feydy A, Barat M et al (2023) Artificial intelligence in diagnostic and interventional radiology: where are we now? Diagn Interv Imaging 104:1–5. https://doi.org/10.1016/j.diii.2022.11.004
24. Fazal MI, Patel ME, Tye J, Gupta Y (2018) The past, present and future role of artificial intelligence in imaging. Eur J Radiol 105:246–250. https://doi.org/10.1016/j.ejrad.2018.06.020
25. Lee L, Salami RK, Martin H, Shantharam L, Thomas K, Ashworth E et al (2024) How i would like AI used for my imaging": children and young persons' perspectives. Eur Radiol. https://doi.org/10.1007/s00330-024-10839-9
26. Huang J, Shlobin NA, Lam SK, DeCuypere M (2022) Artificial intelligence applications in pediatric brain tumor imaging: a systematic review. World Neurosurg 157:99–105. https://doi.org/10.1016/j.wneu.2021.10.068
27. Sammer MBK, Akbari YS, Barth RA, Blumer SL, Dillman JR, Farmakis SG et al (2023) Use of artificial intelligence in radiology: impact on pediatric patients, a white paper from the ACR pediatric AI workgroup. J Am Coll Radiol 20:730–737. https://doi.org/10.1016/j.jacr.2023.06.003

28. Sammer MBK, Sher AC, Towbin AJ (2022) Ensuring adequate development and appropriate use of artificial intelligence in pediatric medical imaging. AJR Am J Roentgenol 218:182–183. https://doi.org/10.2214/ajr.21.26339
29. Salman R, Nguyen HN, Sher AC, Hallam KA, Seghers VJ, Sammer MBK (2023) Diagnostic performance of artificial intelligence for pediatric pulmonary nodule detection in computed tomography of the chest. Clin Imaging 101:50–55. https://doi.org/10.1016/j.clinimag.2023.05.019
30. Chen IY, Lu AX, Erdman L (2025) Lack of children in public medical imaging data points to growing age bias in biomedical AI. medRxiv 06(06):25328913 https://doi.org/10.1101/2025.06.06.25328913
31. Rengan V, Lalwani D, Bhat S, Sundaram PM (2024 Sep 2) Enhancing dataset quality for AI in radiology: challenges and solutions. J Gastrointest Abdom Radiol [Internet]. [cited 2024 Sep 29]; Available from: >https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0044–1790232
32. Nazer LH, Zatarah R, Waldrip S, Ke JXC, Moukheiber M, Khanna AK et al (2023) Bias in artificial intelligence algorithms and recommendations for mitigation. PLoS Digit Health 2:e0000278. https://doi.org/10.1371/journal.pdig.0000278
33. Tripathi S, Gabriel K, Dheer S, Parajuli A, Augustin AI, Elahi A et al (2023) Understanding biases and disparities in radiology AI datasets: a review. J Am Coll Radiol 20:836–841. https://doi.org/10.1016/j.jacr.2023.06.015
34. Mongan J, Moy L, Kahn CE Jr. (2020) Checklist for artificial intelligence in medical imaging (CLAIM): a guide for authors and reviewers. Radiology: Artificial Intelligence 2:e200029. https://doi.org/10.1148/ryai.2020200029

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